Geometry Area, Surface Area & Volume Cheat Sheet

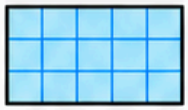
Unit Focus: *area formulas, surface area, volume, and units*

1 AREA AND UNITS

Area

The amount of space inside a two-dimensional figure.

Measured in **square units** (such as cm^2, m^2, in^2, ft^2).

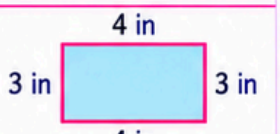


Area = 12 square units

Perimeter

The distance around a two-dimensional figure.

Measured in **linear units** (such as cm, m, in, ft).

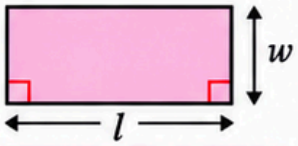


Perimeter = 14 in

2 RECTANGLE AND SQUARE

Rectangle

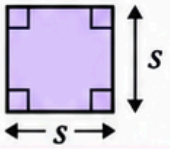
A rectangle has length l and width w.



$$A = lw$$
$$P = 2l + 2w$$

Square

A square has side length s (all sides are equal).

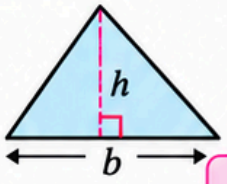


$$A = s^2$$
$$P = 4s$$

3 TRIANGLE AND PARALLELOGRAM

Triangle

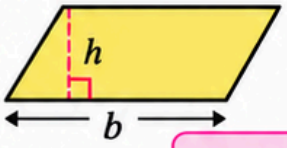
A triangle's area is half the product of the base and height.



$$A = \frac{1}{2}bh$$

Parallelogram

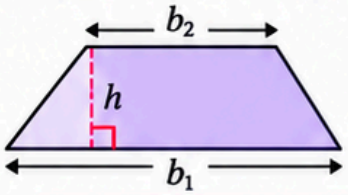
A parallelogram's area is the product of the base and perpendicular height.



$$A = bh$$

4 TRAPEZOID

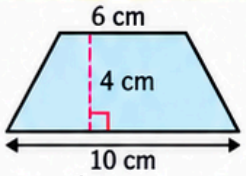
A trapezoid has two parallel bases with lengths b_1 and b_2 and height h.



$$A = \frac{1}{2}(b_1 + b_2)h$$

Example:

Find the area of the trapezoid.

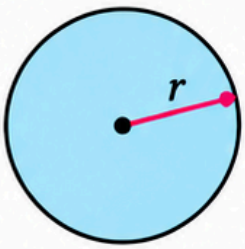


$$A = \frac{1}{2}(6 + 10)(4)$$
$$= \frac{1}{2}(16)(4)$$
$$= 32 \text{ cm}^2$$

5 CIRCLE

A circle's area depends on the radius r, and its circumference (distance around) is also based on the radius.

$$A = \pi r^2$$
$$C = 2\pi r$$



6 COMPOSITE FIGURES

Find the area of composite figures by adding or subtracting the areas of simpler shapes.

Example: Find the area of the L-shaped figure.

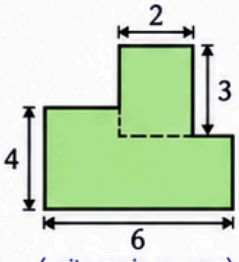
Step 1: Find the area of each rectangle.

$A_1 = 6 \times 4 = 24$

$A_2 = 2 \times 3 = 6$

Step 2: Add the areas.

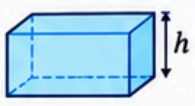
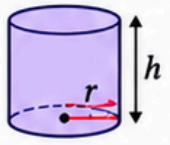
Total area = 24 + 6 = 30 square units



(units are in squares)

7 SURFACE AREA AND VOLUME PREVIEW

Surface area is the sum of the areas of all the faces of a three-dimensional figure.
Volume is the amount of space inside a figure.
Volume is measured in **cubic units** (such as cm^3, m^3, in^3, ft^3).

Figure	Volume Formula	Diagram
Right Rectangular Prism (or any right prism)	$V = Bh$ B = area of the base	
Cylinder	$V = \pi r^2 h$	

Remember:
B is the area of the base, and h is the perpendicular height.

8 QUICK REMINDERS & COMMON MISTAKES

DO REMEMBER:

- Use square units (cm^2, m^2, in^2, ft^2) for area and cubic units (cm^3, m^3, in^3, ft^3) for volume.
- Use the perpendicular height (the shortest distance) when finding area (not a slanted side).
- Keep π in your answer (use π) unless the problem asks you to approximate with a decimal.
- Break composite figures into simpler shapes and add or subtract their areas.

COMMON MISTAKES:

- Using linear units (like cm or m) for area or cubic units for a two-dimensional figure.
- Using a slanted side instead of the perpendicular height.
- Forgetting to square or cube the units.
- Rounding π too early (use π) unless told to approximate).
- Forgetting to add or subtract areas in composite figures.

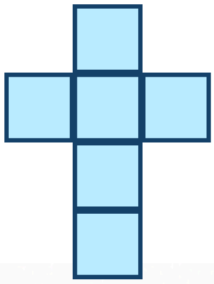
Geometry Area, Surface Area & Volume Cheat Sheet

More Practice: *surface area, volume, and 3D figures*

1 SURFACE AREA BASICS

Surface area = total area of outside faces.

Net of a cube (6 faces)



A cube has 6 congruent square faces.

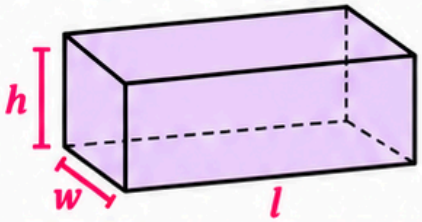
$$SA = 6s^2$$

Remember:

- Surface area is measured in square units.
- Find the area of each face and add them together.

2 PRISMS

A **prism** has two congruent, parallel bases and rectangular lateral faces. A **right** prism lateral faces that are rectangles.



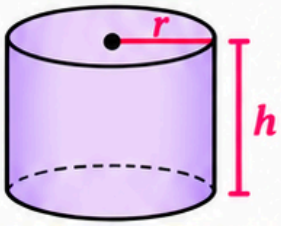
$$SA = Ph + 2B$$
$$V = Bh$$

P = perimeter of the base
B = area of the base
h = height of the prism

Example: A rectangular prism with length 5, width 3, and height 4.

$SA = 2(lw + lh + wh) = 2(5 \cdot 3 + 5 \cdot 4 + 3 \cdot 4)$
$= 2(15 + 20 + 12) = 94$ square units
$V = lwh = 5 \cdot 3 \cdot 4 = 60$ cubic units

3 CYLINDER



$$SA = 2\pi r^2 + 2\pi rh$$
$$V = \pi r^2 h$$

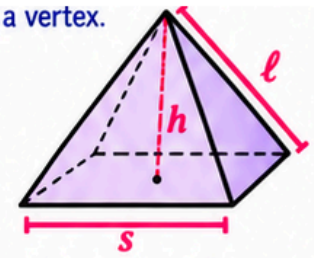
Example: A cylinder with radius $r = 3$ and height $h = 8$.

$SA = 2\pi(3)^2 + 2\pi(3)(8) = 18\pi + 48\pi = 66\pi$ (square units)
$V = \pi(3)^2(8) = 72\pi$ (cubic units)

4 PYRAMIDS

A regular pyramid has a polygon base and triangular lateral faces.

at a vertex.



(square pyramid)

$$SA = B + \frac{1}{2}P\ell$$
$$V = \frac{1}{3}Bh$$

B = area of the base
P = perimeter of the base
ℓ = slant height (height of a lateral face)
h = vertical height

5 CONE AND SPHERE

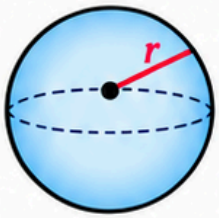
Cone



$$SA = \pi r^2 + \pi r\ell$$
$$V = \frac{1}{3}\pi r^2 h$$

r = radius (base)
h = vertical height
ℓ = slant height

Sphere

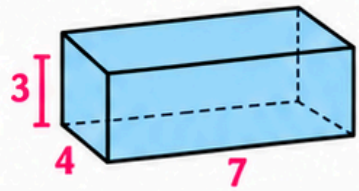


$$SA = 4\pi r^2$$
$$V = \frac{4}{3}\pi r^3$$

r = radius (same for all directions)

6 SURFACE AREA EXAMPLE

Find the surface area of a rectangular prism with length 7, width 4, and height 3.



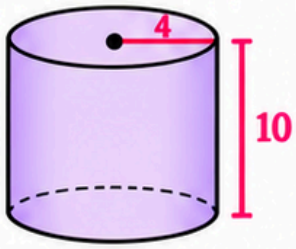
Solution:

$SA = 2(lw + lh + wh)$
$= 2(7 \cdot 4 + 7 \cdot 3 + 4 \cdot 3)$
$= 2(28 + 21 + 12)$
$= 2(61) = 122$

The surface area is 122 square units.

7 VOLUME EXAMPLE

Find the volume of a cylinder with radius 4 and height 10.



Solution:

$V = \pi r^2 h$
$= \pi(4)^2(10)$
$= 16\pi(10)$
$= 160\pi$

The volume is 160π cubic units.

8 QUICK REMINDERS & COMMON MISTAKES

DO REMEMBERS:

- Surface area is the sum of the areas of all outside faces.
- Surface area uses square units (e.g., cm^2, m^2).
- Volume uses cubic units (e.g., cm^3, m^3).
- The slant height (ℓ) is not the same as the vertical height (h).
- Keep π in your answer unless the problem asks you to approximate.
- B represents the area of the base in the formulas.

COMMON MISTAKES:

- Forgetting to include all faces when finding surface area.
- Using square units for volume (or cubic units for area).
- Using the vertical height instead of the slant height in pyramid or cone surface area formulas.
- Dropping π or rounding too early.
- Confusing B (area of base) with P (perimeter of base).
- Forgetting that a cylinder has two circular bases and a curved lateral surface.